EE 66 Fall 2026 — Signals, Dynamics, and Information — Discussion 8B

1. Short Proofs

Let's define $\mathbf{S} = \mathbf{A}^\top \mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is an arbitrary real $m \times n$ matrix. $\mathbf{V} = \begin{bmatrix} \mathbf{v}_1 \cdots \mathbf{v}_n \end{bmatrix}$ is the matrix of normalized eigenvectors of $\mathbf{S}$ with eigenvalues $\lambda_1, \cdots, \lambda_n$.

(a) Let $\mathbf{v_1}, \cdots, \mathbf{v_r}$ be those normalized eigenvectors that correspond to non-zero eigenvalues (i.e. $\|\mathbf{v}_i\| = 1$ and $\lambda_i \neq 0$ for $i = 1, \cdots, r$). Show that $\|\mathbf{A}\mathbf{v}_i\|^2 = \lambda_i$.

Answer: We know that $\mathbf{v}_i$ is an eigenvector of $\mathbf{S}$, so we can start with:

$$\mathbf{S}\mathbf{v}_i = \lambda_i \mathbf{v}_i$$
$$\mathbf{v}_i^\top \mathbf{S}\mathbf{v}_i = \lambda_i \mathbf{v}_i^\top \mathbf{v}_i.$$
$$\mathbf{v}_i^\top \mathbf{S}\mathbf{v}_i = \lambda_i \|\mathbf{v}_i\|^2.$$

On the second step, we multiplied both sides with $\mathbf{v}_i^\top$. Since $\|\mathbf{v}_i\|^2 = 1$ the right hand side will simplify to λ_i. By substituting $\mathbf{S} = \mathbf{A}^\top \mathbf{A}$:

$$\mathbf{v}_i^\top \mathbf{A}^\top \mathbf{A}\mathbf{v}_i = \lambda_i$$
$$(\mathbf{A}\mathbf{v}_i)^\top \mathbf{A}\mathbf{v}_i = \lambda_i$$
$$\|\mathbf{A}\mathbf{v}_i\|^2 = \lambda_i.$$

(b) Following the assumptions in part (a), show that $\mathbf{A}\mathbf{v}_i$ is orthogonal to $\mathbf{A}\mathbf{v}_j$.

Hint: It may help to recall the Spectral Theorem.

Answer: To show that two vectors are orthogonal we have to take the inner product and show that it is zero:

$$(\mathbf{A}\mathbf{v}_i)^\top \mathbf{A}\mathbf{v}_j = \mathbf{v}_i^\top \mathbf{A}^\top \mathbf{A}\mathbf{v}_j = \mathbf{v}_i^\top \mathbf{S}\mathbf{v}_j.$$

We know that $\mathbf{S}\mathbf{v}_j = \lambda_j \mathbf{v}_j$. Therefore,

$$\mathbf{v}_i^\top \mathbf{S}\mathbf{v}_j = \lambda_j \underbrace{\mathbf{v}_i^\top \mathbf{v}_j}_{=0 (\text{since } \mathbf{v}_i \perp \mathbf{v}_j)} = 0.$$

(c) Show that if $\mathbf{V} \in \mathbb{R}^{n \times n}$ is an orthonormal square matrix, then $||\mathbf{x}||^2 = ||\mathbf{V}\mathbf{x}||^2$ for all $\mathbf{x} \in \mathbb{R}^n$.

Hint: Write the norm as an inner product instead of trying to do this elementwise.

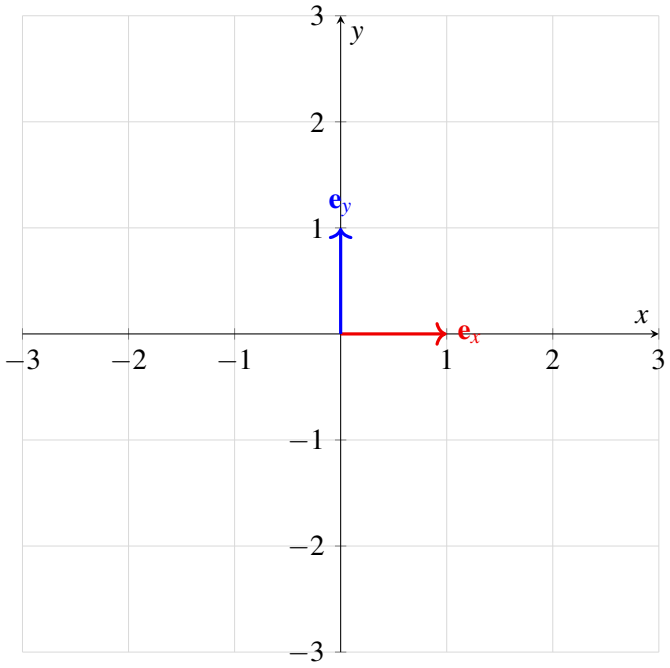
Answer:

$$||\mathbf{V}\mathbf{x}||^2 = \langle \mathbf{V}\mathbf{x}, \mathbf{V}\mathbf{x} \rangle = \mathbf{x}^\top \mathbf{V}^\top \mathbf{V}\mathbf{x} = \mathbf{x}^\top \mathbf{I}_{n \times n} \mathbf{x} = ||\mathbf{x}||^2.$$

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2. Geometric Interpretation of the SVD

(a) When we plot the transformation given by a specific matrix, we think about how the matrix transforms the standard basis vectors. In 2D, let $\mathbf{e}_x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. The vectors $\mathbf{e}_x$ and $\mathbf{e}_y$ are shown below

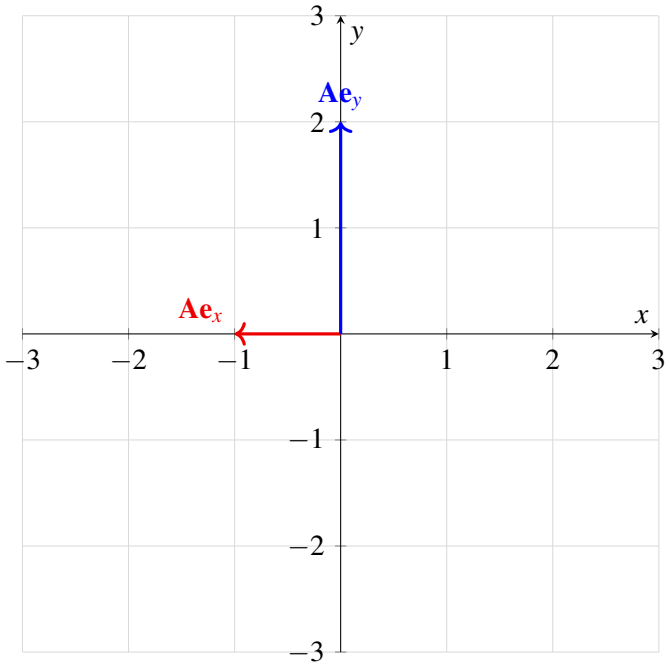


Consider the following matrix

$$\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$$

How would A transform $\mathbf{e}_x$ and $\mathbf{e}_y$? Plot the result.

Answer: We have that $\mathbf{A}\mathbf{e}_x = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ and $\mathbf{A}\mathbf{e}_y = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$. Plotting these, we have



(b) Let's take a look at the 2×2 rotation matrix. Recall that this matrix will rotate any vector counter-clockwise by θ.

$$\mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{1}$$

Show that this matrix is orthonormal.

Answer: We can also show orthonormality by showing that the columns have unit norm and that they are orthogonal. We can also show that this matrix is orthonormal by showing that $\mathbf{R}^\top \mathbf{R} = \mathbf{I}_{2\times 2}$.

$$\|\mathbf{r_1}\| = \left\| \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \right\| = \sqrt{\cos^2\theta + \sin^2\theta} = 1 \tag{2}$$

$$\|\mathbf{r_2}\| = \left\| \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix} \right\| = \sqrt{\sin^2\theta + \cos^2\theta} = 1 \tag{3}$$

$$\langle \mathbf{r_1}, \mathbf{r_2} \rangle = \begin{bmatrix} -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} = -\cos\theta\sin\theta + \cos\theta\sin\theta = 0 \tag{4}$$

(c) Now let's consider how this transformation looks in the lens of the SVD. You are given the following matrix $\mathbf{A}$:

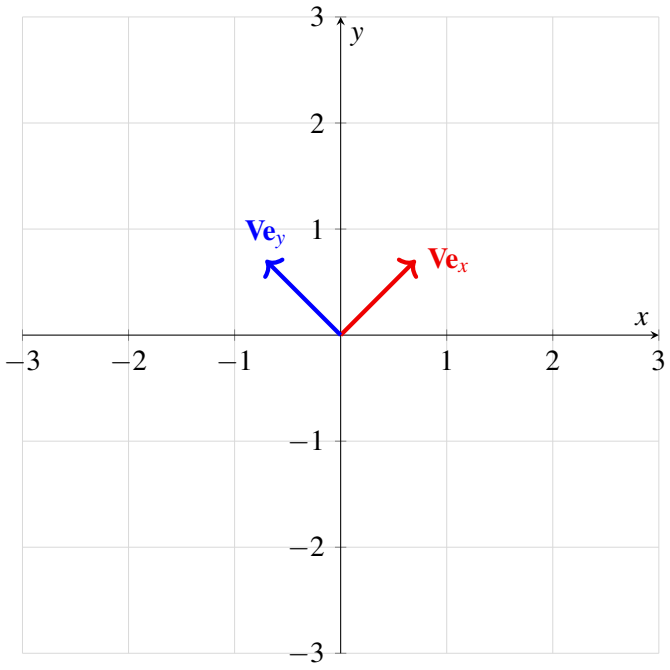
$$\mathbf{A} = \begin{bmatrix} -1 & -2 \\ 2 & 1 \end{bmatrix} \tag{5}$$

Recall that the SVD of this matrix is given by $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$. Assume you are told that

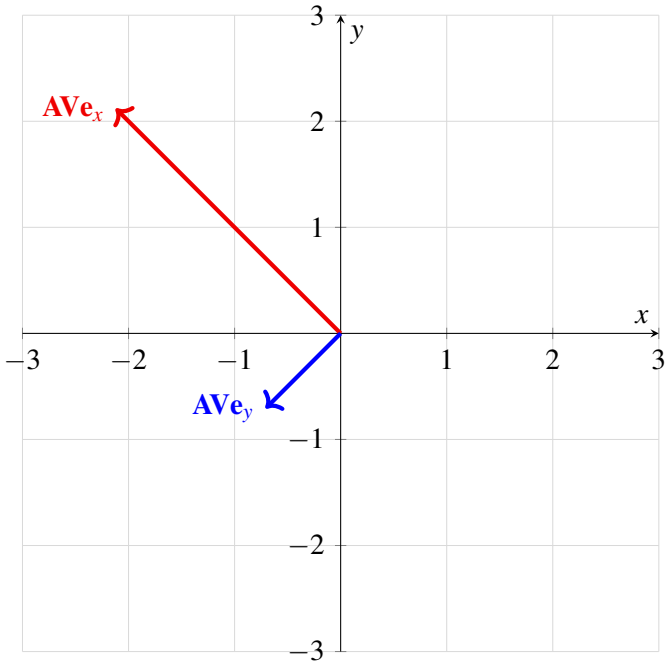
$$\mathbf{V} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \tag{6}$$

We will try to deduce $\mathbf{U}$ and Σ graphically, and the confirm our results numerically. **Plot the transformation given by $\mathbf{V}$ by showing how it affects $\mathbf{e}_x$ and $\mathbf{e}_y$ via left multiplication.** *(HINT: Try writing $\mathbf{V}$ as a rotation matrix with a specific θ.)*

Answer: We notice that $\mathbf{V}$ is a rotation matrix with $\theta = \frac{\pi}{4}$. Hence, it will rotate $\mathbf{e}_x$ and $\mathbf{e}_y$ by $\frac{\pi}{4}$ counterclockwise.



(d) Suppose you are told that the transformation of $\mathbf{AV}$ on $\mathbf{e}_x$ and $\mathbf{e}_y$ looks like



Write this transformation $\mathbf{AV}$ in terms of $\mathbf{U}$ and Σ. Recall that the $\mathbf{U}$ matrix is an orthonormal matrix so it will correspond to any rotations or reflections, and the Σ matrix is a diagonal matrix and will perform any scaling operations. **Based on this fact and the plot of the transformation above, write down a guess for what $\mathbf{U}$ and Σ might be.**

Answer: We notice that $\mathbf{AV} = \mathbf{U}\Sigma$ by right multiplying our SVD by $\mathbf{V}$. Now, it is reasonable to assume that, since $\mathbf{AVe}_x$ appears 3 times as long as $\mathbf{AVe}_y$, then $\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$. Furthermore, it appears as if the vectors have been rotated by $\frac{3\pi}{4}$ so it is likely that $\mathbf{U}$ is a rotation matrix with $\theta = \frac{3\pi}{4}$, i.e., $\mathbf{U} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$.

(e) **Based on the given $\mathbf{V}$ matrix, compute the SVD.** Does your answer match your hypothesis from the previous part?

Answer: We can compute Σ and $\mathbf{U}$ as follows:

$$\mathbf{Av}_1 = \sigma_1 \mathbf{u}_1 \tag{7}$$

$$\mathbf{Av}_2 = \sigma_2 \mathbf{u}_2 \tag{8}$$

More explicitly,

$$\mathbf{Av}_1 = \begin{bmatrix} -1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{3}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} \end{bmatrix} \tag{9}$$

$$\mathbf{Av}_2 = \begin{bmatrix} -1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \tag{10}$$

We can set $\sigma_1 = \|\mathbf{A}\mathbf{v}_1\| = 3$ and $\sigma_2 = \|\mathbf{A}\mathbf{v}_2\| = 1$. These choices yield $\mathbf{u}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ and $\mathbf{u}_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$. Hence,

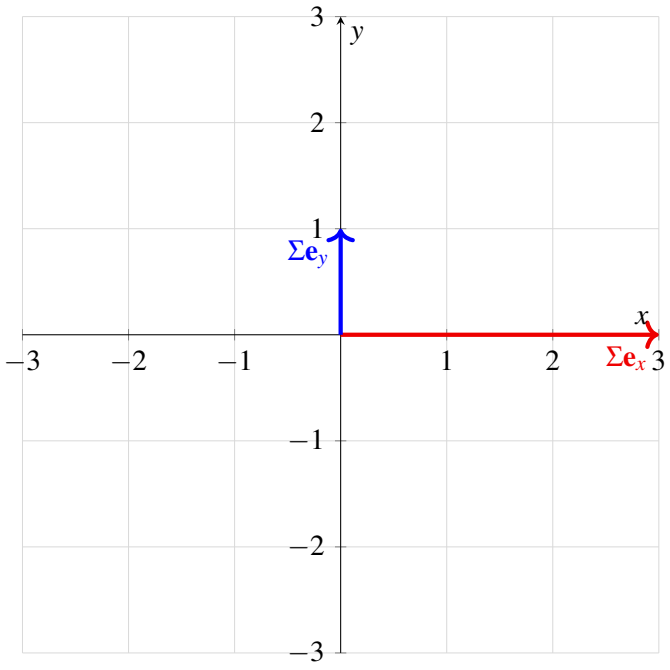
$$\Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} \tag{11}$$

$$\mathbf{U} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \tag{12}$$

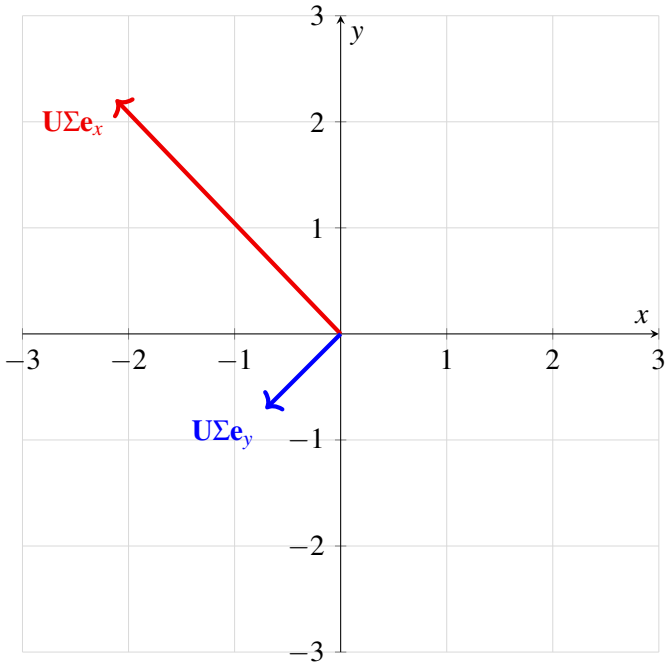
We notice that $\mathbf{U}$ is a rotation matrix with $\theta = \frac{3\pi}{4}$, and indeed, this matches with the $\mathbf{U}$ and Σ we hypothesized in the previous part.

(f) **Using your answer for $\mathbf{U}$ and Σ from the previous part, plot the transformation of Σ on $\mathbf{e}_x$ and $\mathbf{e}_y$. From here, plot the transformation of $\mathbf{U}\Sigma$ on $\mathbf{e}_x$ and $\mathbf{e}_y$.** Does the final plot resemble the transformation shown by $\mathbf{AV}$?

Answer: We notice that $\Sigma\mathbf{e}_x = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$ and $\Sigma\mathbf{e}_y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Hence, we get the following plot:



Now, we noticed that $\mathbf{U}$ is a rotation matrix with $\theta = \frac{3\pi}{4}$ so this will rotate the graph above by $\frac{3\pi}{4}$. This yields



which exactly matches what was given above.